

Exploring Superconducting Quantum Circuits

An introduction to different tools

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Abstract

We present here different theoretical tools to analyze superconducting circuits, notably Energy Participation Ratio (EPR) and Blackbox quantization. We present how EPR manifests in classical mechanics, and we use it to analytically calculate EPR and Blackbox quantization of a transmon in a 3D cavity. We prove that both the approaches are equivalent, and an approach can be chosen based on the problem. Then we introduce the current mirror qubit and present its the wave equation and decoherence analysis.

Introduction

Superconducting circuits are multi-mode systems by construction, where Josephson junctions interact with complex electromagnetic environments defined by cavities, transmission lines, and circuit modes. Traditional lumped-element models are often insufficient for capturing these interactions. Black-box quantization provides a powerful framework for deriving effective Hamiltonians directly from the simulated or measured impedance of a device, bypassing the need for oversimplified models. However, as systems become larger and more coupled, simply extracting mode spectra is not enough, and it also becomes computationally heavy.

Here, the Energy Participation Ratio (EPR) method is used. This quantifies how much each non-linear element participates in each normal mode, allowing an accurate calculation of anharmonicities, dispersive shifts, and coherence properties. Together, EPR and Black-box quantization provides a route to design and optimize superconducting quantum devices, bridging the gap between electromagnetic simulation and circuit Hamiltonians.

The current mirror qubit, a precursor to the commonly studied $0 - \pi$ qubit, is known as an intrinsically protected superconducting qubit. Performing analysis on this will help us understand how topological aspects of a qubit will play a role in qubit properties, especially decoherence.

Introduction to EPR and Blackbox quantization

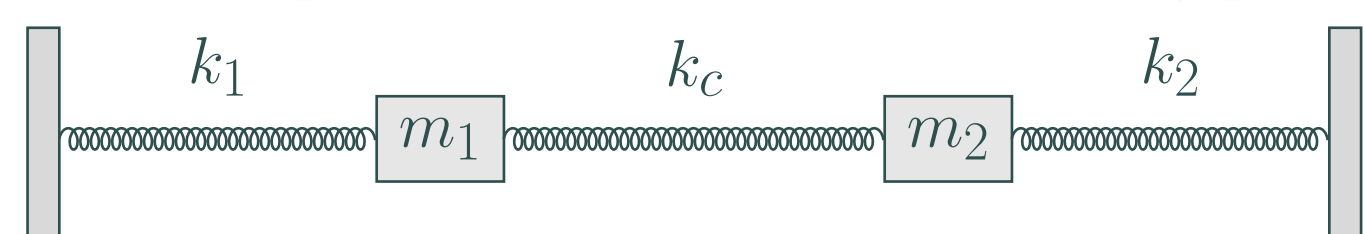
Blackbox quantization: As qubit complexities increase, the coupling strength increases and the models derived from cavity QED struggle to incorporate this. So, the Black box quantization technique was introduced to address these issues. The goal of Blackbox quantization is to find a basis that incorporates the main effect of coupling and perturbatively accounts for weak anharmonicity, assuming that charge dispersion is neglected.

Energy Participation Ratio: The energy participation ratio, introduced by Minev et. al is a useful tool for analyzing circuits. It provides a way to rewrite the Kerr terms in terms of quantities that are easier to find. It also eliminates the need of finite element sweep, and instead relies on much simpler eigenmode analysis.

$$p_m = \frac{\text{Inductive energy present in the Josephson Junction}}{\text{Total inductive energy stored in the mode m}}$$

EPR and it's analogy to classical mechanics

We take the simple system of a coupled mechanical oscillator as a toy problem to understand EPR.



The Lagrangian can be written as

$$\mathcal{L} = \frac{1}{2}m_1\dot{q}_1^2 + \frac{1}{2}m_2\dot{q}_2^2 - \frac{1}{2}k_1q_1^2 - \frac{1}{2}k_2q_2^2 - \frac{1}{2}k_c(q_1 - q_2)^2 \quad (1)$$

$$= \frac{1}{2}\dot{\vec{Q}}^T M \dot{\vec{Q}} - \frac{1}{2}\vec{Q}^T K \vec{Q} \quad (2)$$

$$= \frac{1}{2}\dot{\vec{Q}}^T \tilde{Q} - \frac{1}{2}\vec{Q}^T \Omega^2 \vec{Q} \quad (3)$$

where,

$$Q = E\tilde{Q} \quad (4)$$

$$E = U^T M^{1/2} = \frac{1}{\sqrt{1+R_z^2}} \begin{pmatrix} \sqrt{m_1} & R_z\sqrt{m_2} \\ -R_z\sqrt{m_1} & \sqrt{m_2} \end{pmatrix}, \quad (5)$$

$$U = \frac{1}{\sqrt{1+R_z^2}} \begin{bmatrix} 1 & -R_z \\ R_z & 1 \end{bmatrix}, R_z = \frac{m_2(k_1 + k_c) - m_1(k_2 + k_c) + \sqrt{A^2 + 4m_1m_2B}}{2k_c\sqrt{m_1m_2}} \quad (6)$$

and Ω^2 is the matrix containing square of eigenfrequencies as its diagonal elements. We define participation ratio as $p_{mn} = u_{mn}^2$.

Results (Transmon qubit)

We apply these techniques both analytically and numerically on a transmon in a 3-D rectangular cavity.

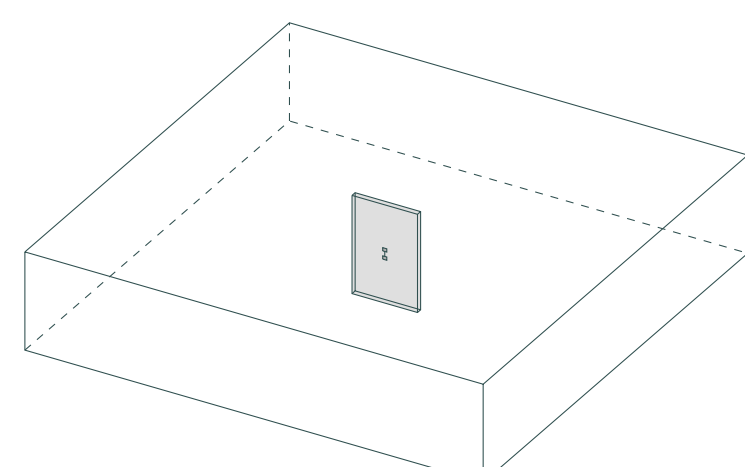


Figure 1: Schematic diagram of transmon in a 3D rectangular cavity

Analytic calculation:

Using the analogy from the previous case, the EPR matrix can be written as

$$\Pi = \begin{bmatrix} \frac{D+A}{2D} & \frac{B^2}{2D(D+A)} \\ \frac{B^2}{2D(D+A)} & \frac{D+A}{2D} \end{bmatrix} = \begin{bmatrix} p_1 & p_2 \\ p_2 & p_1 \end{bmatrix}, \quad (7)$$

where

$$A = L_r(C_r + C_c) - L_J(C_q + C_c) \quad B = 2C_c\sqrt{L_rL_J},$$

$$D = \sqrt{(L_r(C_r + C_c) - L_J(C_q + C_c))^2 + 4C_c^2L_rL_J}.$$

We can rewrite the blackbox expression in terms of EPR as

$$ImY(\omega) = \frac{\left(\frac{\omega}{\omega_1^2} - \frac{1}{\omega}\right)\left(\frac{\omega}{\omega_2^2} - \frac{1}{\omega}\right)}{L_J\omega\left(\frac{p_2}{\omega_1^2} + \frac{p_1}{\omega_2^2}\right) - \frac{L_J}{\omega}} \quad (8)$$

Calculations done numerically using COMSOL verifies the same. We can use EPR and blackbox to find out the self-Kerr and cross-Kerr of the modes.

Current Mirror Qubit

The current mirror qubit is an array of Josephson junctions, $2N$ of them, but then there is a twist in the end, like a Möbius strip. The upper and lower stripes are connected by capacitors, thus forming N units, as seen in the figure below.

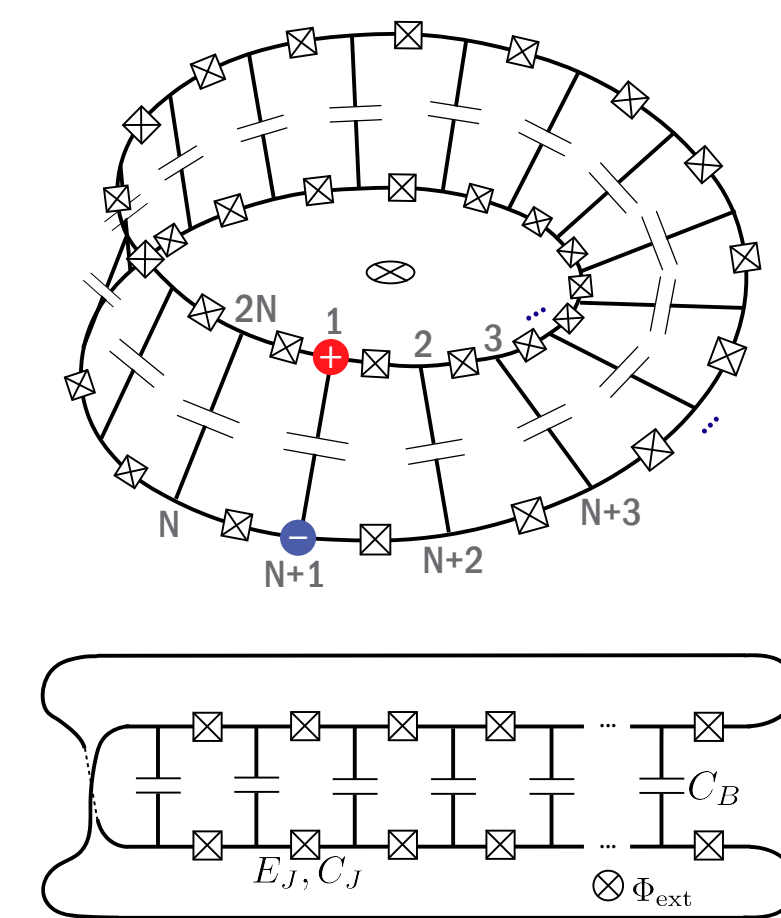


Figure 2: The current mirror qubit. Image taken from [4]

We have the Lagrangian

$$\mathcal{L} = \frac{1}{2} \left(\frac{\phi_0}{2\pi} \right)^2 \sum_{i,j=1}^{2N} \dot{\phi}_i C_{ij} \dot{\phi}_j + \hbar \sum_{j=1}^{2N} \dot{\phi}_j n_{gj} + \sum_{j=1}^{2N} E_J \cos \left(\phi_{j+1} - \phi_j - \frac{\phi_{ext}}{2N} \right), \quad (9)$$

where C_{ij} is given by

$$C_{ij} = \begin{cases} C_g + 2C_J + C_B, & i = j \\ -C_J, & i = j \pm 1 \\ -C_B, & i = j \pm N \\ 0, & \text{otherwise} \end{cases}$$

$$C_{eff} = C_g + 4C_B$$

We obtain the wave equation by finding the equation of motion from the Lagrangian as

$$\frac{1}{2} \left(\frac{\phi_0}{2\pi} \right)^2 \left[C_{eff} \ddot{\phi} - C_J a^2 \frac{d^2}{dt^2} \left(\frac{\partial^2 \phi}{\partial x^2} \right) \right] + \hbar n_g(x) = -2E_J \sin \left(a^2 \frac{\partial^2 \phi}{\partial x^2} \right) \cos \left(2a \frac{\partial \phi}{\partial x} - \frac{\phi_{ext}}{N} \right) \quad (10)$$

DMRG analysis of this qubit provides a sweet spot for the external flux (ϕ_{ext}/ϕ_0) to be 0 or 2π , where it has a degeneracy at 0 and π . The gap between the degeneracy reduces as we increase ϕ_{ext} but this comes with the cost of reducing decoherence.

Conclusions

- EPR and Blackbox observe the same quantities but take different approaches.
- The optimal approach depends on the problem (More number of Josephson junctions might make blackbox difficult).
- We can use this to analyse complex qubit systems, like those with multiple josephson junctions.
- The current mirror qubit provides an idea to construct a qubit having inherent topological protection to noises.

Forthcoming Research

We are trying to apply this techniques to qubits of opposite anharmonicities, which will cancel out the first order nonlinearities. We are also trying apply these tools to the current mirror qubit, which promises topological protection.

References

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